

EPA Air Emissions Programs

(POLL_RPT_COMBINED_EMISSIONS)

The POLL_RPT_COMBINED_EMISSIONS dataset combines facility-level emissions from four EPA programs. This overview states only facts confirmed against EPA documentation or verified directly in the data (sources at the end).

The four programs

- **NEI — National Emissions Inventory.** EPA’s National Emissions Inventory is “released every three years” (triennial) and is a comprehensive estimate of air emissions of “criteria pollutants, criteria precursors, and hazardous air pollutants” from air-emission sources.
- **CAMD — Clean Air Markets / Acid Rain Program.** The Acid Rain Program requires major reductions of sulfur dioxide and nitrogen oxide emissions from electric utilities through a “cap and trade” approach, with affected sources required to install continuous emissions monitors that report measured (not estimated) emissions.
- **GHGRP — Greenhouse Gas Reporting Program.** EPA reports that approximately 8,000 facilities are required to report their greenhouse-gas emissions annually.
- **TRI — Toxics Release Inventory.** Established by the Emergency Planning and Community Right-to-Know Act (EPCRA, 1986; Section 313 of EPCRA established the TRI). EPA reports that the TRI program currently covers 810 individually listed chemicals and 34 chemical categories. TRI is a separate statute from the Clean Air Act.

Fields

Columns of POLL_RPT_COMBINED_EMISSIONS.csv (verified against the file header):

- **REPORTING_YEAR** — calendar year of the emission report.
- **REGISTRY_ID** — Facility Registry Service (FRS) ID; the join key to ICIS-Air.
- **PGM_SYS_ACRNM** — source program: EIS (NEI), E-GGRT (GHGRP), TRIS (TRI), CAMDBS (CAMD).
- **PGM_SYS_ID** — program-specific facility identifier.
- **POLLUTANT_NAME** — name of the reported pollutant.
- **ANNUAL_EMISSION** — emission value for the facility, pollutant, and year.
- **UNIT_OF_MEASURE** — Pounds or MTCO2e (metric tons CO₂-equivalent).
- **NEI_TYPE** — pollutant category: CAP (criteria air pollutant), GHG (greenhouse gas), HAP (hazardous air pollutant), OTH, PFAS.

- NEI_HAP_VOC_FLAG — flag identifying a hazardous-air-pollutant volatile organic compound.

Considerations

- **Mixed units.** Greenhouse-gas values are in metric tons CO₂-equivalent; other values are in pounds. The two units cannot be summed without conversion.
- **Different reporting universes and frequencies.** NEI is triennial while CAMD, GHGRP, and TRI report annually, and each program has its own reporting criteria, so a facility may appear in one program but not another and coverage differs by program and year.

Sources

- EPA, *National Emissions Inventory (NEI)*, <https://www.epa.gov/air-emissions-inventories/national-emissions-inventory-nei>
- EPA, *Greenhouse Gas Reporting Program*, <https://www.epa.gov/ghgreporting>
- EPA, *What is the Toxics Release Inventory?*, <https://www.epa.gov/toxics-release-inventory-tri-program/what-toxics-release-inventory>
- EPA, *AFS Data Download* documentation (April 2014) — Acid Rain Program description.
- Direct verification against `data/raw/POLL_RPT_COMBINED_EMISSIONS.csv`.